

# Infraestructura Real Desplegada — SRBRGDOCSAIPROD

**Estado:** Verificado contra pipelines (azure-pipelines.yml, azure-pipelines-functions.yml, azure-pipelines-admin.yml)

**Fecha de fotografía de infraestructura:** 2026-06-30

**Región:** swedencentral (upe48)

**Resource Group:** SRBRGDOCSAIPROD

## Topología Completa

SRBRGDOCSAIPROD (Resource Group)

- Azure Functions
  - Name: srbappprodocai
  - Runtime: .NET 10 isolated
  - Managed Identity: Enabled
  - Storage: AzureWebJobsStorage (from KeyVault)
- App Service (Admin Blazor)
  - Name: srbwebadminprodocai
  - Runtime: .NET 8
  - Managed Identity: Enabled + RBAC to KeyVault
  - Access: Key Vault via "Key Vault Secrets User" role
- App Service (AssetResolver Plugin)
  - Name: srbwebpluginassetresolver
  - Runtime: .NET 8
  - Connection: AssetResolver DB (secret: user-ods-dwh)
  - Base URL: https://srbwebpluginassetresolver.azurewebsites.net/
- Key Vault
  - Name: srbkvprodocai
  - Secrets (40+): SQL, Storage, Extraction creds, Classification creds, GDC creds, API keys
  - RBAC: Functions App + Admin App + AssetResolver can read
- SQL Server Database
  - Connection: SqlConnectionString (from KeyVault)
  - Migrations: EF Core 8 packages; applied manually/on-demand (dotnet-ef CLI 9.x)
  - Schema: DocumentIADbContext (12 entities)
- Azure Storage (Blob + Table)
  - Connection: AzureWebJobsStorage + AzureStorageConnectionString
  - Purpose: Durable Functions hub, document storage, audit logs
  - Expiration: Automated via FechaExpiracionBlob index
- Application Insights
  - Endpoint: Implicit via APPINSIGHTS\_INSTRUMENTATIONKEY (from KeyVault)
  - Tracing: PromptTracing enabled (20,000 char limit per prompt)
  - Sampling: 20 events/sec (from local.host.json)
- Azure AI Services
  - Azure Content Understanding (CU)
    - Endpoint: https://upe48-mm2avmdm-swedencentral.services.ai.azure.com/
    - API Key: from KeyVault
    - Processing Location: geography
  - Document Intelligence (DI)
    - Endpoint: https://srbdiprodocai.cognitiveservices.azure.com/
    - API Key: from KeyVault
    - API Version: 2024-11-30
  - OpenAI (GPT Fallback)
    - Endpoint: https://upe48-mm2avmdm-swedencentral.openai.azure.com
    - Model: gpt-4o-mini
    - Deployment: gpt-4o-mini
    - API Key: from KeyVault

## Configuración Post-Despliegue (App Settings)

### Azure Functions (srbappprodocai)

| Setting        | Value           | Source       |
|----------------|-----------------|--------------|
| Runtime        | dotnet-isolated | hardcoded    |
| Secrets Source | AzureVault      | hardcoded    |
| KeyVault Name  | srbkvprodocai   | pipeline var |

Extraction Configuration

| Setting                     | Value                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| DefaultProvider             | azure-content-understanding                                                                                                           |
| CU Endpoint                 | <a href="https://upe48-mm2avmdm-swedencentral.services.ai.azure.com/">https://upe48-mm2avmdm-swedencentral.services.ai.azure.com/</a> |
| CU MaxConcurrentCalls       | 4 (pipeline var, not in host.json)                                                                                                    |
| CU HardTimeout              | 90 seconds                                                                                                                            |
| CU CircuitBreaker           | enabled                                                                                                                               |
| CU CircuitBreaker Threshold | 5 failures                                                                                                                            |
| CU CircuitBreaker Open Time | 45 seconds                                                                                                                            |
| CU Max Retries              | 3                                                                                                                                     |
| CU Retry Delay              | 500ms                                                                                                                                 |
| GPT Fallback Enabled        | true                                                                                                                                  |
| GPT Fallback Endpoint       | <a href="https://upe48-mm2avmdm-swedencentral.openai.azure.com">https://upe48-mm2avmdm-swedencentral.openai.azure.com</a>             |
| GPT Fallback Model          | gpt-4o-mini                                                                                                                           |
| GPT Fallback MinFieldsRatio | 0.9                                                                                                                                   |
| GPT Fallback Timeout        | 60 seconds                                                                                                                            |
| GPT Fallback Temperature    | 0.0                                                                                                                                   |
| GPT Fallback MaxTokens      | 2000                                                                                                                                  |

Classification Configuration

| Setting                  | Value                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|
| DefaultProvider          | azure-document-intelligence                                                                                               |
| DefaultModelKey          | default.azure-di                                                                                                          |
| DI Endpoint              | <a href="https://srbdiprodocai.cognitiveservices.azure.com/">https://srbdiprodocai.cognitiveservices.azure.com/</a>       |
| DI API Version           | 2024-11-30                                                                                                                |
| GPT Fallback Enabled     | true                                                                                                                      |
| GPT Fallback Endpoint    | <a href="https://upe48-mm2avmdm-swedencentral.openai.azure.com">https://upe48-mm2avmdm-swedencentral.openai.azure.com</a> |
| GPT Fallback Model       | gpt-4o-mini                                                                                                               |
| GPT Fallback Threshold   | 0.5                                                                                                                       |
| GPT Fallback Timeout     | 30 seconds                                                                                                                |
| GPT Fallback Temperature | 0.0                                                                                                                       |
| GPT Fallback MaxTokens   | 150                                                                                                                       |

GDC (Legacy Document Management System)

| Setting               | Value                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Endpoint              | <a href="https://srbwidd03.sareb.srb:8090/sintws/IDocService">https://srbwidd03.sareb.srb:8090/sintws/IDocService</a> |
| HTTP Basic Auth       | Username + Password (from KeyVault)                                                                                   |
| Application ID        | CKP1                                                                                                                  |
| Document Type ID      | document                                                                                                              |
| Content Field Name    | Content                                                                                                               |
| Origen Documento      | 8878                                                                                                                  |
| Clase Expediente      | AI04                                                                                                                  |
| Default Matricula     | AI-99-SCXX-00                                                                                                         |
| Servicer              | 9999                                                                                                                  |
| Entidad Origen        | 9999                                                                                                                  |
| Proceso Carga         | PC01                                                                                                                  |
| Tipo Expediente       | AI                                                                                                                    |
| Publico               | verdadero                                                                                                             |
| SSL Validation Bypass | true                                                                                                                  |
| Timeout               | 60 seconds                                                                                                            |

Prompt Tracing

| Setting               | Value  |
|-----------------------|--------|
| Enabled               | true   |
| Include Prompt Text   | true   |
| Max Prompt Text Chars | 20,000 |

Storage & Database

| Setting                      | Value                    |
|------------------------------|--------------------------|
| AzureWebJobsStorage          | @Microsoft.KeyVault(...) |
| AzureStorageConnectionString | @Microsoft.KeyVault(...) |
| SqlConnectionString          | @Microsoft.KeyVault(...) |
| Run DB Migrations on Startup | false                    |

Asset Resolver Integration

| Setting               | Value                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| AssetResolver BaseUrl | <a href="https://srbwebpluginassetresolver.azurewebsites.net/">https://srbwebpluginassetresolver.azurewebsites.net/</a> |
| AssetResolver ApiKey  | @Microsoft.KeyVault(...)                                                                                                |

App Service — Admin (srbwebadminprodocai)

| Setting                       | Value                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|
| FunctionsAdminApi_BaseUrl     | <a href="https://srbappprodocai.azurewebsites.net/api/">https://srbappprodocai.azurewebsites.net/api/</a> |
| FunctionsAdminApi_FunctionKey | @Microsoft.KeyVault(VaultName=srbkvprodocai;SecretName=FunctionsAdminApiFunctionKey)                      |

**RBAC:** Managed Identity has "Key Vault Secrets User" role.

**App Service — AssetResolver (srbwebpluginassetresolver)**

| Setting                            | Value                                                                |
|------------------------------------|----------------------------------------------------------------------|
| ConnectionStrings__AssetResolverDb | @Microsoft.KeyVault(VaultName=srbkvprodocai;SecretName=user-ods-dwh) |
| ApiKey                             | @Microsoft.KeyVault(...)                                             |

**Proceso de Despliegue Real**

**Pipeline:** azure-pipelines.yml (manual trigger)

**Stage 1: Build & Test**

1. Use .NET 10 SDK
  2. Restore solution
  3. Build solution (Release)
  4. Run unit tests (from DocumentIA.Tests.Unit)
  5. Publish 3 projects:
    - DocumentIA.Functions
    - DocumentIA.Admin
    - DocumentIA.AssetResolver
  6. Upload 3 artifacts to pipeline cache

**Stage 2: Run Database Migrations (conditional: currently disabled)**

1. Install dotnet-ef global tool (v9)
  2. Restore solution
  3. Build solution
  4. Apply EF Core migrations:
    - Project: src/backend/DocumentIA.Data/
    - Startup: src/backend/DocumentIA.Functions/
    - Connection: SqlConnectionString (from KeyVault)

**Stage 3: Deploy Functions**

1. Download drop-functions artifact
  2. Deploy to srbappprodocai using zipDeploy
  3. Apply 40+ App Settings (post-deploy validation)
  4. Validate all CU resiliencia variables are applied

**Stage 4: Deploy Admin**

1. Download drop-admin artifact
  2. Deploy to srbwebadminprodocai using zipDeploy
  3. Assign Managed Identity to Admin App
  4. Create RBAC role: Key Vault Secrets User
  5. Apply Admin App Settings (Functions API endpoint + key)

**Stage 5: Deploy AssetResolver**

1. Download drop-assetresolver artifact
2. Deploy to srbwebpluginassetresolver using zipDeploy
3. Apply AssetResolver App Settings (DB connection, API key)

### Stage 6: Validate Configuration Contract

1. Run validate-azure-appsettings-contract.ps1
2. Verify all required settings are present on 3 apps
3. Fail deployment if contract not met

## Key Deployment Characteristics

### Security

- Managed Identities for all 3 apps
- RBAC (Key Vault Secrets User)
- No connection strings in code — all from KeyVault
- GDC endpoint has SSL bypass (legacy system workaround)

### Concurrency Control

- **CU MaxConcurrentCalls:** 4 (set via App Settings post-deploy)
- **DF Activity Concurrency:** 4 (set via host.json, not configurable)
- **DF Orchestrator Concurrency:** 4 (hardcoded in durable functions runtime)

### Timeouts

| Service                       | Timeout         | Configurable     |
|-------------------------------|-----------------|------------------|
| CU (Content Understanding)    | 90s             | via App Settings |
| GPT (Extraction fallback)     | 60s             | via App Settings |
| DI (Document Intelligence)    | 120s (inferred) | via App Settings |
| GPT (Classification fallback) | 30s             | via App Settings |
| GDC                           | 60s             | via App Settings |

### Resilience

- **Circuit Breaker:** Enabled for CU (threshold=5, open=45s)
- **Retries:** CU retries 3x with 500ms initial delay
- **Fallback Chain:** CU → GPT (extraction), DI → GPT (classification)

### Observability

- Application Insights integrated (via APPINSIGHTS\_INSTRUMENTATIONKEY)
- Prompt tracing enabled (20,000 char limit per prompt)
- Custom metrics from PromptTracing configuration
- Sampling: 20 events/sec (configured in local host.json)

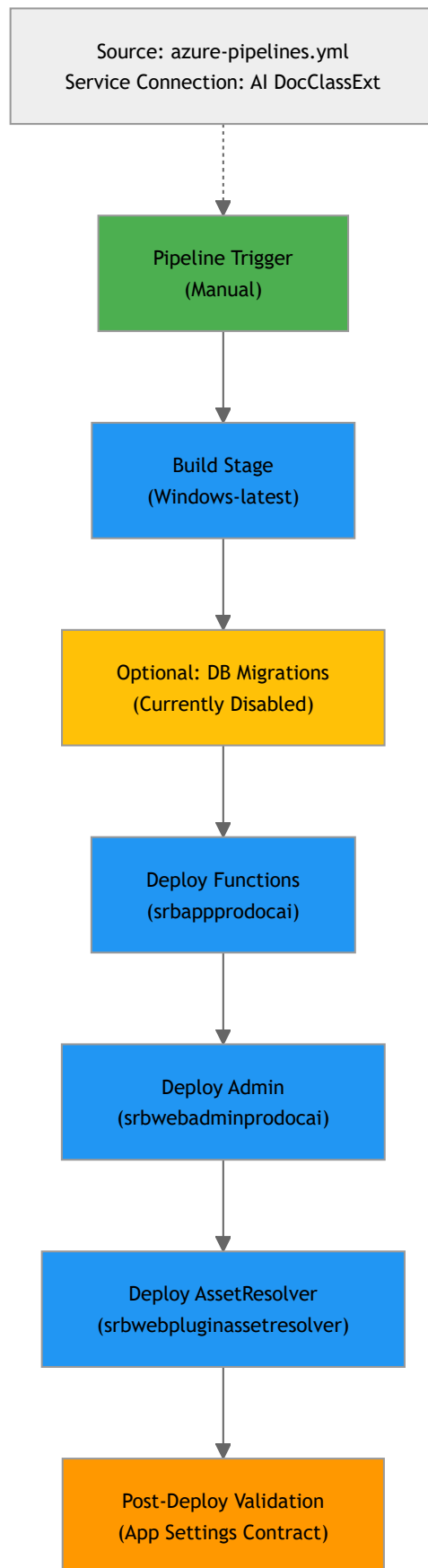
## Key Vault Secrets (Referenced in Pipelines)

| Secret Name                                       | Usage                       | Environment |
|---------------------------------------------------|-----------------------------|-------------|
| SqlConnectionString                               | Database connection         | Production  |
| AzureWebJobsStorage                               | Durable Functions hub       | Production  |
| AzureStorageConnectionString                      | Document + blob storage     | Production  |
| Extraction--AzureContentUnderstanding--ApiKey     | CU authentication           | Production  |
| Extraction--GptFallback--ApiKey                   | GPT extraction fallback     | Production  |
| Classification--AzureDocumentIntelligence--ApiKey | DI authentication           | Production  |
| Classification--GptFallback--ApiKey               | GPT classification fallback | Production  |
| GDC--HttpBasicUsername                            | GDC legacy system auth      | Production  |
| GDC--HttpBasicPassword                            | GDC legacy system auth      | Production  |
| AssetResolverApiKey                               | AssetResolver plugin        | Production  |
| FunctionsAdminApiFunctionKey                      | Admin app ↔ Functions auth  | Production  |
| user-ods-dwh                                      | AssetResolver database      | Production  |
| APPINSIGHTS_INSTRUMENTATIONKEY                    | App Insights (inferred)     | Production  |

## Endpoints Reference

| Component            | Endpoint                                                                                                                              | Status     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------|
| Functions API        | <a href="https://srbappprodoci.azurewebsites.net/api/">https://srbappprodoci.azurewebsites.net/api/</a>                               | Production |
| Admin Blazor Web     | <a href="https://srbwebadminprodoci.azurewebsites.net/">https://srbwebadminprodoci.azurewebsites.net/</a>                             | Production |
| AssetResolver Plugin | <a href="https://srbwebpluginassetresolver.azurewebsites.net/">https://srbwebpluginassetresolver.azurewebsites.net/</a>               | Production |
| CU Service           | <a href="https://upe48-mm2avmdm-swedencentral.services.ai.azure.com/">https://upe48-mm2avmdm-swedencentral.services.ai.azure.com/</a> | Production |
| DI Service           | <a href="https://srbdiprodoci.cognitiveservices.azure.com/">https://srbdiprodoci.cognitiveservices.azure.com/</a>                     | Production |
| OpenAI Service       | <a href="https://upe48-mm2avmdm-swedencentral.openai.azure.com">https://upe48-mm2avmdm-swedencentral.openai.azure.com</a>             | Production |
| GDC Legacy           | <a href="https://srbwidd03.sareb.srb:8090/sintws/IDocService">https://srbwidd03.sareb.srb:8090/sintws/IDocService</a>                 | Production |

## Deployment Orchestration



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## Known Issues & Workarounds

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| Issue                       | Current State               | Mitigation                                                          |
|-----------------------------|-----------------------------|---------------------------------------------------------------------|
| GDC SSL Bypass              | BypassSslValidation=true    | Legacy system requirement; consider certificate management          |
| DB Migrations Stage         | Disabled (condition: false) | Migrations must run manually before pipeline or on first deployment |
| Extended Sessions (Durable) | Disabled in host.json       | Documented in EXTENSIBILIDAD; concurrency is hardcoded at 4         |
| No Disaster Recovery        | Not configured              | Backups via Azure automatic retention; no multi-region failover     |

## Notes for Operators

1. **Pipeline Execution:** Manual trigger only. No automatic CI/CD on push.
2. **Deployment Order:** Functions → Admin → AssetResolver (enforced via dependsOn).
3. **Rollback:** Manual via Azure Portal or revert last deployment artifact.
4. **Configuration Changes:** Edit App Settings directly in Azure Portal or update pipeline variables and re-run.
5. **Database Schema Changes:** Apply EF Core migrations manually or enable RunDatabaseMigrationsOnStartup for automatic application.
6. **Key Rotation:** Update secrets in Key Vault; apps read on next restart or configuration refresh.

## Related Documentation

- [DATA\\_MODELS\\_ER\\_DIAGRAM.md](#) — Database schema
- [EXTENSIBILIDAD\\_PLUGIN\\_SYSTEM.md](#) — Provider architecture
- [PERFORMANCE\\_TUNING.md](#) — Concurrency & timeout tuning
- [INDEX.md](#) — Master documentation index

**Estado:** Verificado contra las definiciones de pipeline (azure-pipelines.yml).